

THE POLITICS OF INTERNATIONAL STANDARDS

France and the Color TV War

RHONDA J. CRANE

COMMUNICATION AND INFORMATION SCIENCE

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MELVIN J. VOIGT

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Communication Standards and the Politics of Protectionism

INTRODUCTION

Communication between nations forms the foundation upon which international understanding is built. Countries need the ability to communicate with one another, but the mere possession of technical systems does not ensure that transmission and reception is possible between them. Compatible standards for the technical systems are necessary before communication can take place.

Competing national interests frequently intervene in efforts to establish compatible standards for communication systems and thus make international agreement unfeasible. While incompatibility can be surmounted in some instances by technological developments, circumstances created by an inability to communicate may have severe consequences.

Visualize what could happen if an airplane equipped with microwave landing gear were forced to make an emergency landing at a foreign airport with an incompatible system, and the pilot could not communicate with the control tower (Air traffic control, 1965; Robertson, 1976; Technology: A bitter dispute, 1975). Imagine the economic repercussions if conditions required the urgent computer transmission of bulk financial data to banks in different countries, and incompatible standards prevented the terminals from receiving the information. Or, on a different level, consider the inconvenience, expense, and delays for broadcasters when the exchange of programs between nations operating on different standards necessitates elaborate conversion procedures (Franklin, 1966; Weller, 1966).

If these are examples of problems which can arise when standards for technical systems are incompatible, what prevent nations from adopting compatible standards at the outset?

A basic dilemma underlies the development of international telecommunications, a dilemma nations encounter each time they are forced to choose between standards for different technical systems of communication. The focal point of the dilemma for each nation is the determination of the relative advantages or disadvantages of standards compatibility: Do the advantages of compatibility and the ability to communicate with other nations outweigh the advantages of incompatibility, the ability to protect domestic industries, or control broadcasts or information transmitted across national airways?

In the following chapters, some of the facets of this dilemma are explored in an analysis of the failure to establish internationally compatible technical standards for color television systems. Through a case study of France, the problems involved in the international selection of a European standard for color television systems are examined.

Three different color television systems (developed by the United States, France, and West Germany) were in competition to become the European standard. All three color television systems were incompatible with one another; broadcasts transmitted on one system could not be received on television sets manufactured for a different system. Therefore, there was international interest in the adoption of a single color television standard for all of Europe. The competition between the systems to become the European standard was politically intense; the nation whose system was selected as the European standard would gain large economic revenues and political prestige.

In order to fulfill their ambitions to develop a color television industry and create an export market for the products, the French sought to make their system the European standard. The French were interested in adopting compatible color television standards with the rest of Europe, on condition that the French system would be chosen as the European standard so that France could recoup financial investments from patent and license royalties. If another system were chosen as the European standard, the French could ensure the development of their color television industry with different standards, by creating a different market for the products in Communist, Third World, and less developed countries. The difference in technical standards between the three systems also could be employed as a nontariff barrier to protect the French color television industry during its developmental stages.

Three major themes are investigated.

First, what are the problems that are generated by having differences in national technical standards in communications? Are differences in technical standards effective tools to create an industry?

Second, when and how may a state successfully intervene to support the development of an industry in the private sector? What are some of the problems a government encounters in the promotion of an industry?

Finally, how do political and economic interests influence the course of decision-making in international organizations? What are some of the difficulties confronted by international organizations in attempting to reach international agreements on technological issues?

A discussion of the role of technology in politics is important to understanding the three themes to be explored.

TECHNOLOGICAL POLITICS

Technology is conventionally regarded as a force in world politics, affecting international relations through its diffusion, potential applications, and regulation. Technological developments occurring in one nation may have a profound impact on other nations on the other side of the globe (e.g., Holland, 1968, p. 37; Humphrey, 1968, pp. 68-69; Westfall, 1966, pp. 64-69; Wingert, 1966, pp. 67-69). This interactive effect is likely not only to continue but to expand. With increasing frequency, new technologies are emerging which are global in nature and require the participation or cooperation of many countries, if the full scope of their benefits is to be realized (Skolnikoff, 1967, pp. 302-316; Wallenstein, 1974, pp. 33-38).

In the field of communications, satellites offer the best illustration of this growing trend, with their potential for global use in meteorology, forestry, agriculture, geography, resource mapping, communications, direct broadcasting, data gathering and distribution, navigation and traffic control (Skolnikoff, 1972, pp. 56-62).

The growing applications and continuing advances of technologies such as these are generating important new demands and creating new constraints on the independence of national action. There has been a steady decrease in a nation's freedom of action to apply science and technology as it so chooses, even within its own borders (Skolnikoff, 1967, p. 303).

These pressures are evident in the increasing number of international conflicts over the possession and deployment of technological developments. For many reasons, economic, political, and

military, among others, perhaps all technologies have been affected to some degree. Nations have become increasingly aware and more vocal about the technological business of their neighbors. The technological developments of one nation are objects of intense scrutiny by others. When one nation develops an important technology, rivalry develops among other nations to duplicate or improve it.

Increased international concern over technological developments has led to changes in the assessment of a nation's power. Since World War II, scientific and technological developments have sharply altered established relationships among nations and overturned traditional measures of power and influence (Skolnikoff, 1967, p. 3). A nation no longer measures its strength in military or economic terms alone, but on its technological capacity as well.

One effect of this trend has been an attempt by certain nations to impress other nations with a forceful display of technologies and, in particular, spectacular technologies. The demonstration of one nation's technological abilities requires another nation to surpass these feats, if technological pride and parity are to be maintained. For example, when the Soviet Union launched satellites and put a man into space, the United States retained its technological position by also launching satellites and then placing men on the moon. The Soviets countered by sustaining the duration of men in space, the Americans upheld their position by developing the space shuttle, and so on.

Nations vie with one another to obtain the latest technological developments. Technological strategies vary from heavy governmental support across a broad range of development, to specialization in particular areas of development, to importing another nation's advances, to some combination of the three (Gilpin, 1970). Regardless of the strategy employed to garner technological developments, the underlying assumption remains the same: namely, the greater the show of technological force, the more influence a nation may exert in the international arena. In effect, technological politics have become one of the new battlegrounds for international influence.

In many European countries, the role of technology in the political process was perceived by governments as a means of advancing and retaining political and economic power (Gilpin, 1968). This was one reason why European states intervened to support the development of industries in the private sector. The ramifications of when and how this occurred is one theme explored in this study. A motivating factor was the perceived threat by European countries of an economic invasion from the United States. They feared the potential

colonization of their economy and the concomitant loss of their political independence. Each technological advance identified as American was viewed somewhat suspiciously by European nations which sensed a further setback for themselves in a "technology gap" (Servan-Schreiber, 1967; see also Salomon, 1971; Spencer, 1970).

The French, perhaps more than other Europeans, were particularly sensitive to American technological power. At the international level, France sought to forge a "system of scientific-technological alliances that could effectively balance American power." The goal in France was "to make science an instrument of French economic, military and political objectives" (Gilpin, 1967, p. 7). Industrial development and protectionism were essential elements of this goal.

Although problems might be generated through protectionism (for example, difficulties arising from differences in communication standards with color television systems), the standards could be employed as effective tools to create an industry. While the growth of technological interdependence has led to increasing national and international pressures for countries to cooperate in selecting technologies with compatible technical standards, from the French perspective the advantages of installing incompatible standards were greater. This was particularly the case when the incompatibility could assist in the development of French technological power.

There are many difficulties that have to be overcome in the establishment of internationally compatible technical standards.

THE POLITICS OF SETTING INTERNATIONAL STANDARDS IN COMMUNICATIONS

One problem in setting standards is the question of where the responsibility lies within a country for determining the standards to be selected. This is a problem of national as well as international regulation. There are a variety of national institutional modes of authority for determining standards within a nation, but the increasing technological interdependence of countries has not been matched by the development of international mechanisms with the power to resolve conflicts among them. Consequently, difficulties persist in agreeing upon internationally compatible standards.

The significance of national standards in a commercial industry is well illustrated by the case of color television systems. Once television standards have been adopted by a nation, it is expensive to change them. For any such change, a nation must determine whether

it is politically and economically advisable to scrap all television sets in use, regardless of age, condition, price, etc., because they are incompatible with the proposed new standards. Another consideration is the cost of changing all the professional electronic equipment necessary to transmit on a new system. Can this be accomplished within a political process with many vested interests? Financial pressures often lock nations into the standards they have chosen so that if standards are incompatible initially, they are likely to remain that way.

Trade in the products affected by the differences in standards may be limited to markets which use the same standards. Markets are fragmented by the existence of incompatible standards (Allen, 1966; Standards: Barriers or aids, 1970). This may lessen the economics of scale possible in mass production; it may also lessen the number of potential suppliers for each standard. This is important when the size of the market for a particular technology is taken into account. For example, according to forecasts of the electronics industry, by the 1980s sales of color television sets are expected to be worth \$12 billion annually (Electronic Industries Assn., 1972). In addition, there is the multibillion dollar business in electronics equipment and components associated with the systems. This market could be better exploited by more nations if compatible technical standards existed (Standardization and the development of trade, 1964). Then production costs could be lowered and export opportunities broadened.

Communications problems are created by a lack of compatibility between systems. For example, program exchanges between countries operating on different color television systems become very difficult; elaborate conversion techniques must be employed. This lowers the technical quality of the broadcast, raises program costs, and unnecessarily complicates the exchange. Television reception problems are further exacerbated in some nations possessing linguistic and cultural diversity.

For example, in Belgium or Switzerland, the choice between the French and German standards in color television had to be weighed against the political consequences of satisfying only part of the population (Business international: Colour TV—Belgian cockpit, 1966, pp. 289-90). Ultimately, the choice would play a role in determining whether broadcasts would be received from France or from Germany. When standards are incompatible between neighboring countries, regions in border areas are also affected. Although multistandard sets capable of receiving broadcasts from different standards exist, they are very expensive. Not all viewers affected by incompatible standards are able to afford these sets.

Yet, it is difficult to establish compatible standards. As previously mentioned, there is the political problem of regulation in standard-setting at both the national and international levels. At the national level, the degree of authority within an agency to decide on a particular standard may vary from country to country. The issues of concern for that agency may not always be similar to other domestic political and economic interests. Technical compatibility may be less important to the overall national welfare than the possibility of controlling information transmitted across the country or promoting industrial development. The recommendations of a domestic agency advocating the adoption of compatible standards may be sacrificed in favor of other objectives.

Political and economic interests may negatively influence the outcome of attempts to establish compatible standards when the standards in question are linked to domestically owned patent and license rights. In such situations, the economic rewards from patent and license rights to a technology may be so great that a country simply cannot afford to sacrifice the economic interests of domestic industry in favor of the technical convenience of sharing another nation's standard. Moreover, differences in standards may provide a country with leverage to protect its industry from market invasion from the industry of another country. It is difficult for an industry to recoup the losses from the development of a technology without the royalties from patent and licensing rights, unless it is successful in capturing markets in other countries. Such economic losses may stymie future technological innovation in industry. The difference in standards might be part of that nation's objective in protecting the technology. This would mean no agreement (other than an agreement to adopt its standards) would be possible on compatible standards without a fundamental change in that nation's policies.

It is not a question of adopting an apolitical standard, but a standard that is identified with another country, a standard which symbolizes that nation's power and progress through technology. For one country to be dependent upon another for its technology may be economically as well as psychologically untenable. What happens to the dependent one, if relations between the two become hostile? Are supplies cut? Does the technology involved cease to function because replacement parts are not forthcoming? These factors, described in Chapter 3, influenced decisions by the French government not to be dependent upon American technology during the 1960s.

When a country equates the development and adoption of its standard with its prestige, it may not be politically feasible later to renounce that standard and adopt another one. Consequently, ne-

gotiations may break down and a deadlock occur. This is what happened with the color television systems negotiations in Vienna in 1965 during the meeting of the C.C.I.R. (International Radio Consultative Committee of the I.T.U., International Telecommunications Union) to decide which of the three (German, French, or American) color television systems to adopt as the standard for western Europe.

National political objectives in international relations may also play a role in determining the likelihood of establishing compatible standards. In the international arena, a nation's preference for adopting compatible standards may be perceived as having other important implications. When one nation votes in favor of another nation's standards, the vote may also be interpreted as a signal against a third nation. Many of the votes cast by the countries represented at the 1965 C.C.I.R. conference in Vienna were intended as political signals.

In sum, each nation approaches the issue of compatible standards with a parochial point of view. The determination of whether it is necessary as well as desirable to adopt standards compatible with other nations will be based upon an evaluation of several factors involving the national interest. These include national political strategy, national technical needs, public opinion, estimates of the value of the services, economic status, balance of payments, the costs of not agreeing, history, and experience. Once standards have been adopted, it is virtually impossible to change them.

On the organizational level, the international mechanisms for debate, negotiations, and agreement on standards—especially in communications—lack the authority to impose a solution. One of the principal difficulties in the international standardization of communication technologies arises from the existence of multiple international organizations. Each organization has an interest in a particular aspect of the standardization process. Lack of coordination and poor exchange of information among them sometimes lead to project duplication or support of conflicting standards (U.S. participation in I.E.C., 1966, p. 239).

The formation of regional blocs and regional standardization organizations contributes to this problem. When standards are adopted in regional organizations prior to being considered or recommended by world standardizing bodies, major obstacles to international cooperation are created. Positions become hardened, interests become vested, and change becomes unlikely (U.S. participation in I.E.C., 1966, p. 239). This is particularly true when standards are formulated to erect barriers to importing the goods of nonparticipating nations.

Another set of problems in the international standardization of communication technologies at the organizational level is generated

by the nature of international organizations. International organizations are dependent upon their various members for financial support and technical input (La Que, 1969, p. 145). This is especially troublesome when the international organizations have a voluntary rather than government-affiliated status. In the former instance, success in adopting compatible standards is, to a greater extent, contingent upon the ability of members to submerge insular outlooks and press for cooperation, provided that the selection of compatible standards does not conflict with their national interests.

Agreement on compatible standards in international organizations is also hampered by differences in language (Ainsworth, 1964, p. 365). The need for interpretation facilities raises costs and tends to make the scheduling of frequent meetings improbable. Elaborate procedures within these international organizations are often devised to avoid areas of political sensitivity and bias. For example, committees are formed with as broad a geographical and political representation as possible. In the case of color television systems, each of the eleven subcommittees assigned to test the systems included representatives of different nations in order to prevent the interest of one country from dominating the others. Frequently, members are allowed to state publicly their reservations about particular decisions; issues where an impasse has been reached or where agreement seems unlikely may be dropped. Often, a Byzantine quality exists regarding the nature of negotiating most international compromises (Remley, 1966, p. 44).

Although such efforts are made to maintain as impartial and apolitical an atmosphere as possible within international organizations in the communications field, it is not always possible to do so. For example, in the 1965 Plenipotentiary Conference of the International Telecommunications Union at Montreux, Switzerland, a resolution to expel South Africa from the meeting was introduced by some African nations and upheld by the majority of the participants (New nations show muscle, 1965, p. 71). It is not inconceivable that similar actions might be enforced in the future with regard to Israel, Taiwan, etc. The enormous power accumulated by the Arab nations from oil wealth makes them formidable contenders should they desire to wield their influence in these international organizations.

Another group of problems which thwart negotiations at the organizational level arises from circumstances external to the nature of the international organization. These are problems of a logistical nature, including restrictions on the number of active participants (large working parties of 100 members in some organizations are not unusual, although they tend to make slow progress); time allotted

for drawing up recommendations and clearing disputes (many meetings convene for at least five weeks); and the financing of participants. In the case of the United States, members or their companies must often pay their own way. Influencing decisions regarding standards in international organizations may be very expensive. Members also must take leaves of absence from their companies. Since top technical personnel are frequently involved, this may result in losses to the company (Jones, 1973, p. 207; see also Wallenstein, 1972, p. 370).

There are several factors which influence the success of a technical decision to set compatible standards for a communications technology. The case of color television systems illustrates the difficulties a government faces in the attempt to utilize differences in standards to create an industry. In the following chapters, the political and economic pressures within France which made a decision to adopt another nation's technology, another nation's standard untenable, will be explored. They are typical of the pressures other nations encounter in the dilemma of choosing internationally compatible standards. This book examines how political and economic interests influence the course of decision-making in international organizations, often conflicting with each other and with the goals of establishing compatible technical standards.

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Conclusion

This book has explored the use of technical standards as a non-tariff device for protecting an industry. Many of the same issues arise as in the study of protection by tariffs: Do they work to promote the development of an "infant industry?" Do they raise or lower the productivity of the protected industry? Do they operate in the interest of consumers? Do they restrict international commerce, and if so, what can the international community do about them?

In this case, standards worked as a device for promoting an industry. As one theme of this study illustrated, a state can successfully intervene to support the development of an industry in the private sector. While the success of the development of an industry in the private sector is dependent upon several factors (including industrial capability, technological ownership, economic feasibility, as well as other factors discussed in previous chapters), the ability of a government to exert political control and political influence throughout the process is a key element. In the case of the French color television industry, the government facilitated development from the outset by exercising political control over domestic conflicts of interest, in the electronics industry and in the O.R.T.F., coordinating national resources toward the same goals. One goal was the creation of an export market for color television products, which the French accomplished in part by politically influencing other nations to choose the French color television system (the Franco-Russian accord, the "bonne geste" relationship, etc.).

The results of this study suggest that the chances for a government to succeed in developing an industry in the private sector are increased by limiting opportunities in the development process for factors over which the government has no control or influence. The more control a government exerts over factors related to the production, promotion, sales, and export of a technology, the less likely it is that outside forces can interfere with development plans. A government can minimize intervention opportunities by factors in the development process that cannot be directly controlled (such as regulating the flow of imports by tariffs) by the use of standards.

Second, this research illustrates how standards were employed as nontariff barriers, protecting the French color television industry from German-manufactured color television products. Differences in national technical standards can be effectively utilized to create an industry, but the use of standards as a protective device is limited: a company in one country can always manufacture equipment on the same standard as another company in a foreign country by constructing a plant to manufacture it inside the foreign country (as Radio-technique did in France). Standards may effectively deter trade, but are not an unsurmountable obstacle.

Differences in standards may also be manipulated by governments to segment markets, thereby creating export possibilities where none existed previously. It is a technique for politically controlling the marketplace by technology. Although differences in standards may have beneficial impacts on the development of an industry, they may simultaneously have negative impacts for communications between countries.

In color television systems, the incompatibility between standards was eventually overcome through technological innovation by transcoding devices. What happens when incompatibility between standards cannot be overcome in communications technologies? As technologies of communication proliferate—in radio, telegraph, telephone, computers, etc.—the problems and consequences of the inability to communicate are severe (even sometimes a matter of survival, such as with microwave landing systems at airports). While international cooperation in telecommunications has been fruitful, it has not always been completely successful, as the case of color television systems illustrates. In establishing national telecommunications standards, it is incumbent upon nations to consider the negative effects which internationally incompatible standards might generate. The increasing complexity of communications technologies and the concomitant growth of areas needing standardization raise serious questions about the process in which international standards are set.

The third theme investigated in this study demonstrates how the present process permits national, political, and economic interests to intervene in international standards decisions, frequently causing countries to opt for incompatibility. As the study of color television systems illustrates, royalties and revenues from patent rights are a major hindrance to international agreements. The search for national political prestige and political objectives also create disunity. No mechanisms in international standards-setting impose agreement. Consequently, international negotiations may breakdown, a circumstance which occurred at the Vienna and Oslo meetings.

International organizations are further hampered in their attempts to establish compatible standards by their dependence on member nations for financial support, technical expertise, and willingness to cooperate. However, if technical decisions are concluded by politicians rather than by technical experts, as they were with color television systems, these organizations may atrophy since they no longer serve a useful purpose. Can mechanisms be developed to ensure their independence and technical freedom to set internationally compatible standards? How can the politics of technology be reduced in international standards negotiations?

A first step might be the internationalization of patents for technologies adopted world-wide. All royalties from the patents would belong to the international organization recommending the standards for the technology. These royalties would cover operating expenses of the international organizations, freeing them in part from contributions of member countries. Each country whose technology contributed some original feature to the standard selected would receive a predetermined percentage of the royalties in return; this would enable the country's industry to recoup financial investments in development.

Since the standard chosen would probably be some combination of the many technologies from different nations, the prestige attached to having a national standard adopted world-wide could be eliminated. The international standard could be renamed after the organization recommending its selection, identified and distinguished by a numerical suffix. For example, the standard for color television systems might not have been cited as the American, German, French, or Russian standards, but as C.C.I.R. Standard Number 1965. This would reduce some of the political problems in establishing an international standard by making the standard itself less nationalistic.

Furthermore, nations could be pressured either to adopt compatible standards for communications technologies, or to install a device enabling communications to take place between incompatible

standards. A country would be permitted to retain a standard different from the one internationally recommended, provided that its technology incorporate a mechanism which would guarantee the continual flow of international communications. Perhaps a system of penalties could be enforced against countries refusing to participate. This could be determined by a study on the effectiveness of different penalties utilized internationally against various countries. Penalties might range from international taxes levied on the nonparticipating country to the withholding of key communications materiel imported from participating nations.

Finally, only technical experts who had participated in tests conducted on the different communication systems under consideration would be permitted to negotiate the final agreement and selection of the standard. Politicians would not be accepted as official delegates, and their attendance at these meetings would be prohibited. Political pressure on technical delegates would be reduced by ensuring that the voting results were kept secret and possibly by maintaining no records of an individual delegate's preference.

Why would any nation accept these suggestions for changes in international organizations responsible for setting standards in communications? Primarily because an improvement in the overall process of setting international standards would benefit all nations concerned. An improvement in the system of setting standards would make it more likely that standards would be based on technical criteria. This might eliminate many of the problems generated by competing political and economic interests, apparent in the case of color television systems. Decisions reached in this manner would ensure that international communications would be possible because compatible standards would exist. Furthermore, some economic remuneration would be guaranteed to these nations in which technical contributions had originated, thereby protecting the ability of those industries to recoup financial investments. While recognizing the limitations of these suggestions—including the probability that some nations would never be able to accept them for reasons of national security, competition, hostile relationships, etc.—they are intended as a step toward thinking about changing the international standards-setting process.

International standards-setting in communications technologies could be made more objective, technical, and independent of national interests than currently exists. As long as national, political, and economic interests are permitted to intrude in the decisions, situations of incompatible standards between countries will continue to

occur. What is needed in setting a standard is the facility to choose a system which is technically superior, so that technical decisions are based on technical and not other criteria. Attention should be focused on methods to improve *the process* by which standards are set. In the end, the ability of nations to communicate will depend upon that process.